

Retail Market Basket Intelligence

End-to-End Data Engineering & Association Rule Mining

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Domain: Retail Transaction Data Analytics

Executive Summary

The Core Concept:

Customers build shopping baskets organically; they do not buy items in isolation.

The Methodology:

Market Basket Analysis leverages historical transaction data to decode these hidden product relationships.

The Goal:

Transform raw, chaotic retail data into actionable, data-driven business decisions.

Strategic Objectives



Algorithmic Discovery:

Deploy the Apriori algorithm across 522k+ records to map frequent itemsets.



Behavioral Mapping:

Extract deep insights into how consumers pair products.



Commercial Application:

Directly improve cross-selling, product bundling, and physical store layouts based on mathematical probability.

The Business Challenge

The Challenge

The business possesses massive transactional data, but lacks visibility into true customer purchase patterns.

The Roadblock

Raw data is filled with anomalies (missing IDs, returns, extreme outliers) that distort analysis.

The Solution

Engineer a robust data cleansing pipeline, followed by advanced Association Rule Mining.

Dataset Overview & Schema

DATASET SIZE

522,064 records across 7 features
(Itemname, Quantity, Price,
CustomerID, etc.)

UNIQUE TRANSACTIONS

21,663 distinct retail invoices.

PRODUCT CATALOG

4,186 unique products.

GEOGRAPHIC SCOPE

Spans 30 countries, with a heavy
concentration in the United
Kingdom.

Enterprise Technology Stack



Database Layer

MySQL, SQLAlchemy,
PyMySQL



Data Engineering

Python, Pandas, NumPy



Analytics Engine

mlxtend (Apriori
Machine Learning
Algorithm)



Visual Intelligence

Matplotlib, Seaborn,
NetworkX, Microsoft
Excel

Phase 1 - Exploratory Data Analysis (EDA)



Average Basket Size

24.03 items per transaction (Median: 13).



Average Purchase Quantity

10.09 units per line item.



Customer Concentration

A highly active core customer base (e.g., top user executed 210 individual transactions).



Finding

High variance in purchase behavior indicates a mix of retail and bulk buyers.

Phase 1 - Data Quality Roadblocks



Missing Data

134,041 missing CustomerIDs (25.68%)
hidden as blank strings.



Duplicate Entries

5,286 exact duplicate rows detected.



Logical Anomalies

1,336 negative quantities (representing
returns, not purchases).



Extreme Outliers

Maximum single-item quantity hit
80,995 units, severely skewing normal
consumer metrics.

Phase 2 - The Data Cleansing Pipeline

STEP 1

Dropped duplicates and records with missing product identifiers.

STEP 2

Purged negative quantities and invalid pricing (refunds/freebies).

STEP 3

Applied Interquartile Range (IQR) filtering to remove extreme wholesale outliers.

Result: Dataset optimally reduced from 522,064 to 430,127 high-quality, analysis-ready rows.

Phase 2 - Feature Engineering

Revenue Mapping

Derived total line-item value
(Revenue = Quantity × Price).

Transaction Size

Aggregated by Invoice to
measure exact basket depth.

Product Frequency

Calculated global demand for
every SKU to prepare for
algorithmic filtering.

Phase 2 - Matrix Transformation

The Requirement

Algorithms require binary matrices, not flat transaction tables.



The Pivot

Grouped 17,597 distinct baskets into a massive sparse matrix.



Noise Reduction

Filtered out ultra-rare products (appearing <100 times) to eliminate statistical noise.



Final ML Dataset

A highly optimized 17,597 × 1,295 Boolean matrix.

Phase 3 - Apriori Algorithm Deployment

ALGORITHM DEPLOYED

Apriori (via mlxtend).

MINIMUM SUPPORT

0.02 (Itemset must appear in at least 2% of all transactions).

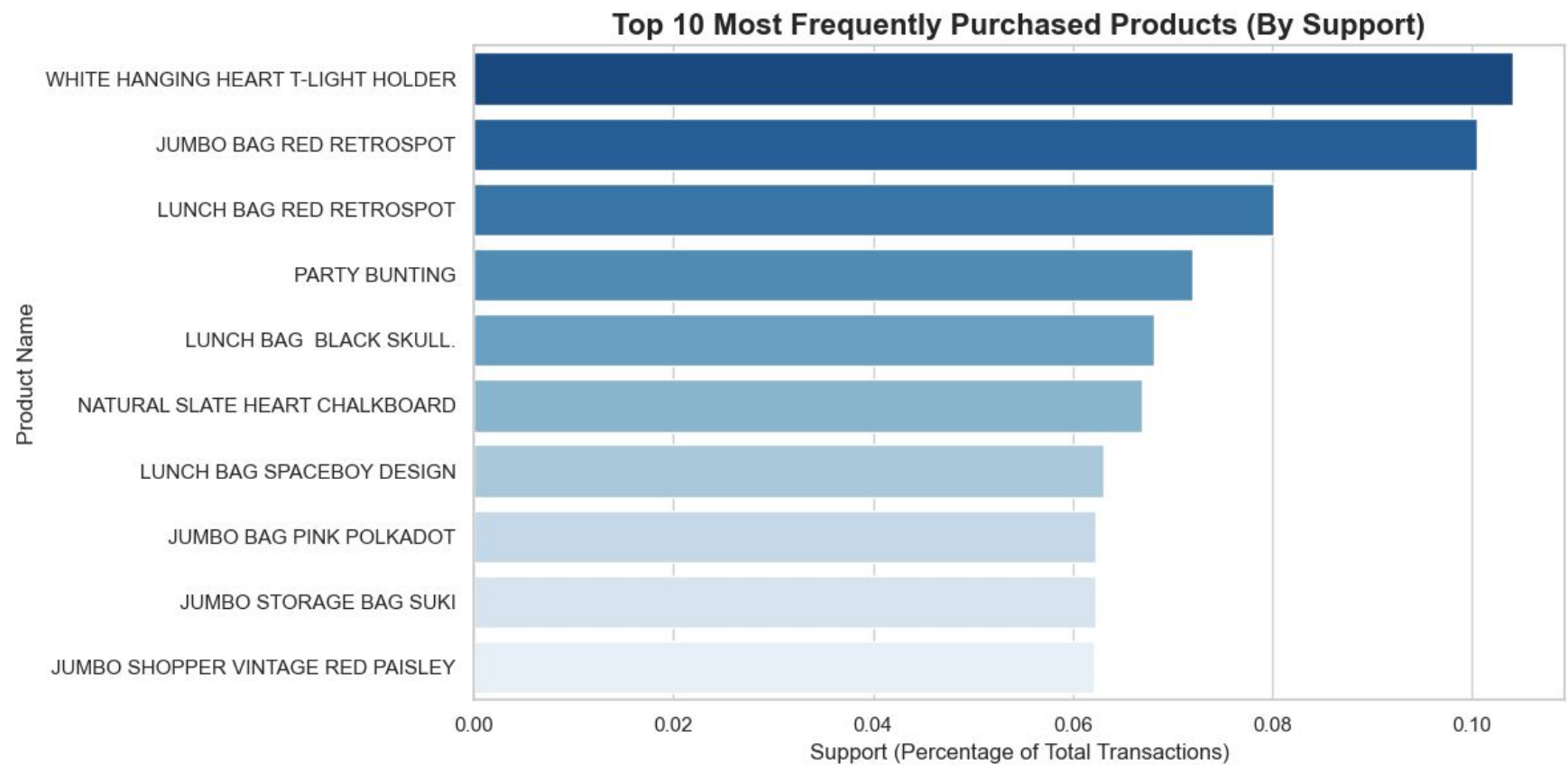
MINIMUM CONFIDENCE

0.50 (At least a 50% probability of cross-sell).

MAXIMUM ITEMSET LENGTH

3 (To ensure rules remain actionable and uncomplicated).

Baseline Demand



Key Takeaway: The WHITE HANGING HEART TLIGHT HOLDER and JUMBO BAG RED RETROSPOT are massive anchor products, each appearing in over 10% of all store transactions.

Defining the Association Rules

Support

How often the product combination happens globally.

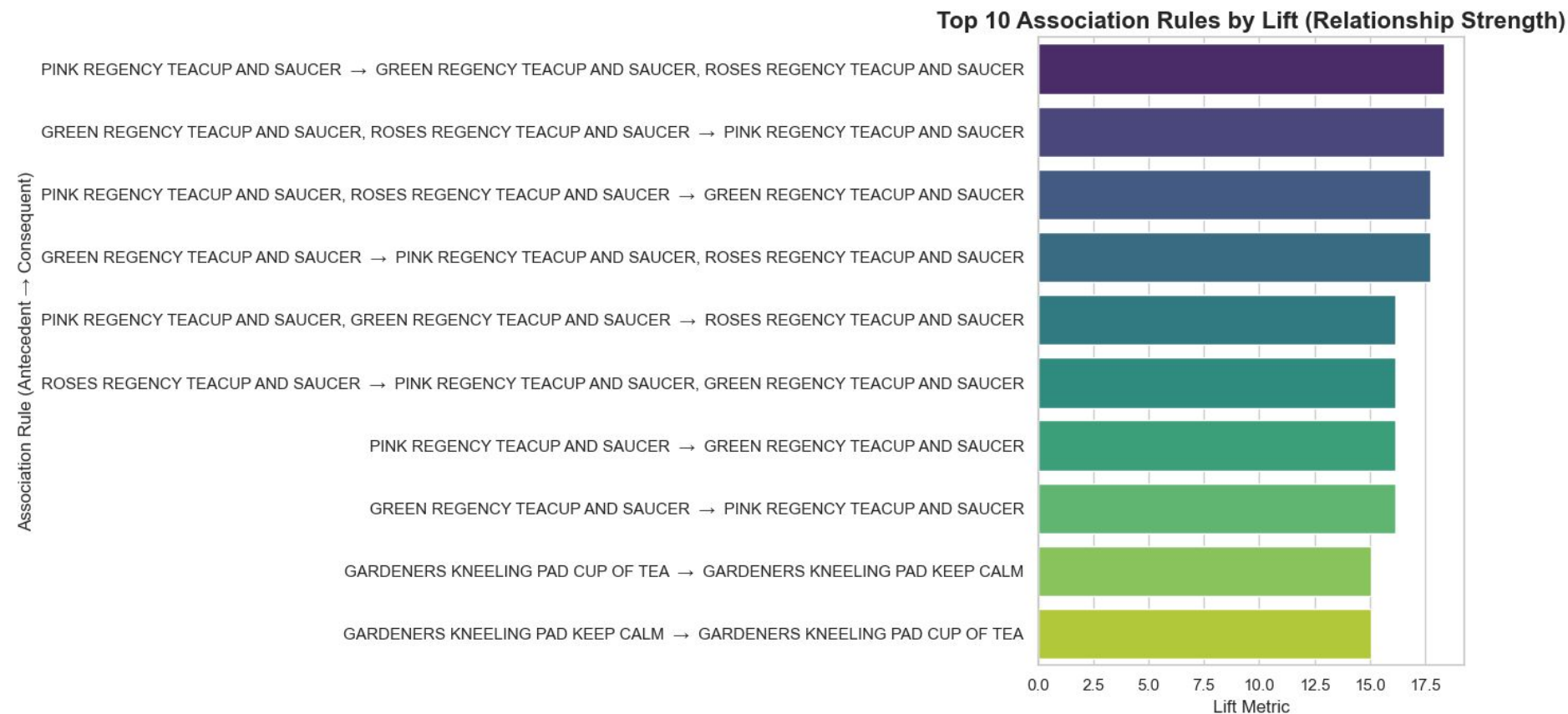
Confidence

The exact probability of the cross-sell occurring.

Lift

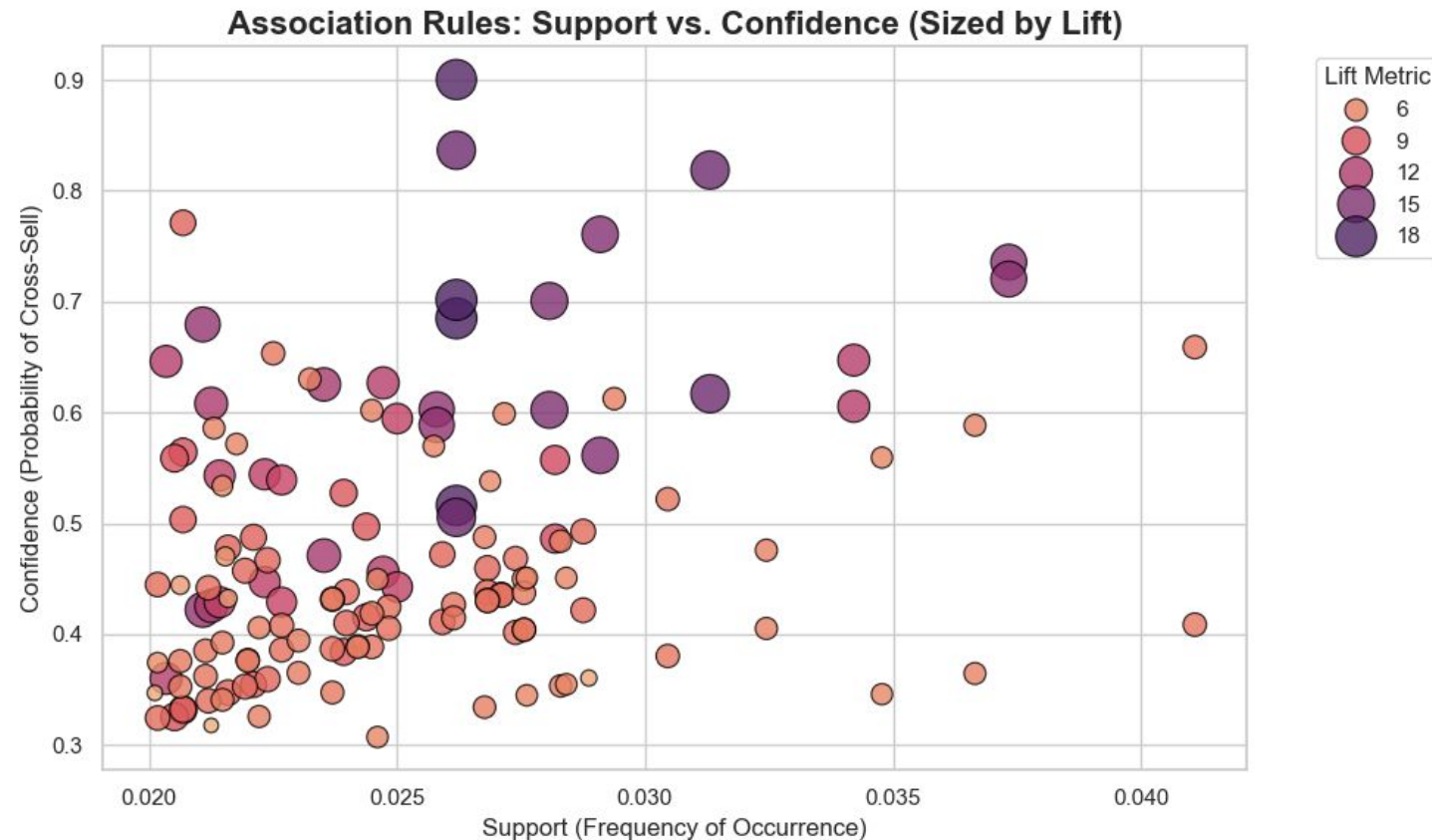
The true strength of the relationship. (Lift > 1 proves the products boost each other's sales beyond random chance).

The Strongest Relationships



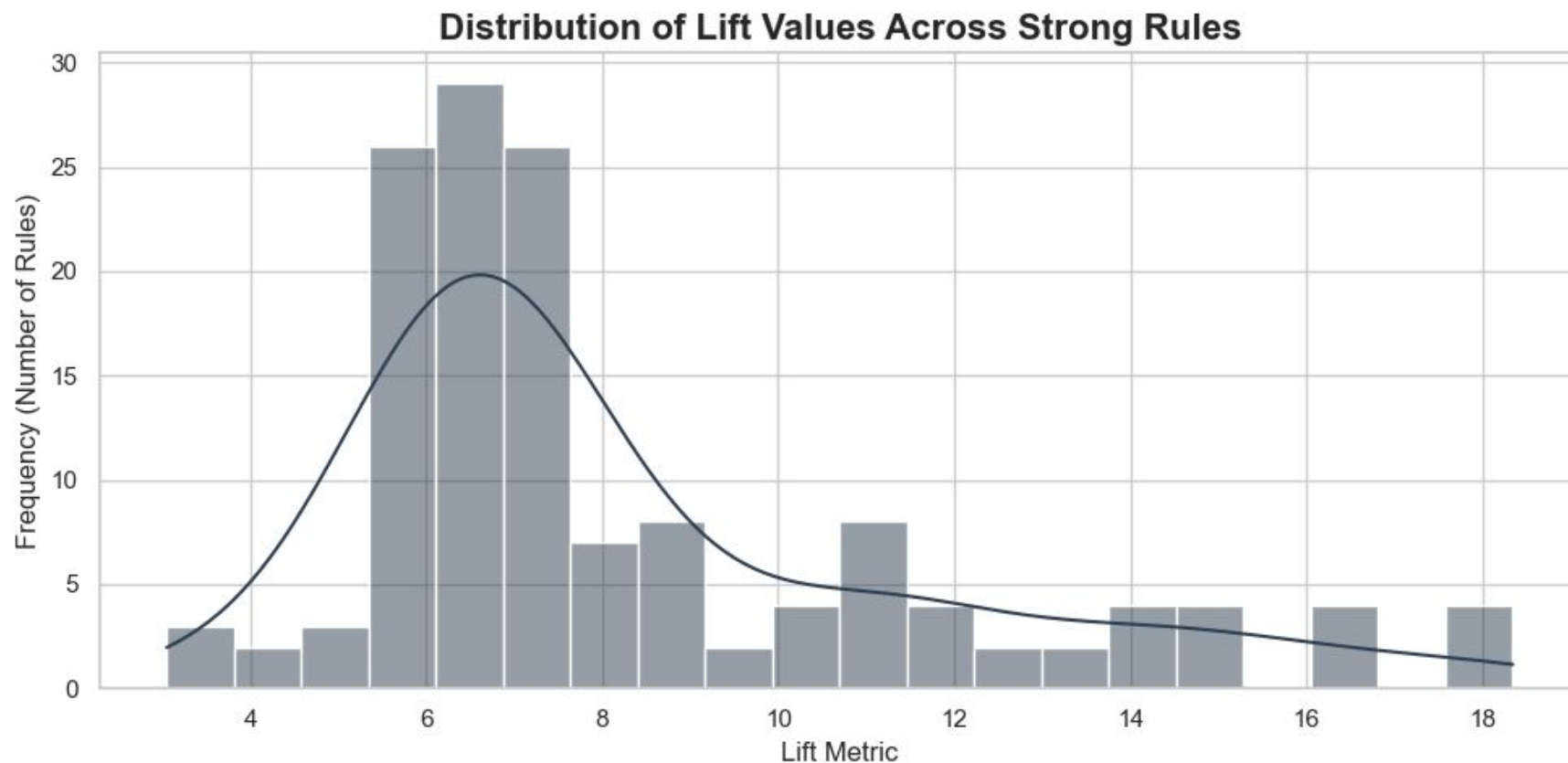
Key Takeaway: The Regency Tea Set combinations dominate the highest-strength relationships, proving customers insist on buying matching sets.

Rule Reliability



Key Takeaway: High clustering in the upper half of the chart proves our extracted rules boast a highly reliable purchase probability (55% to 90% Confidence).

Distribution of Association Strength



Key Takeaway: The vast majority of our rules have a Lift between 5 and 15, confirming exceptional, mathematically significant product pairings.

Discovery of Product Clusters

Regency Tea Sets

Green, Pink, and Roses
teacups. (Exceptional Lift:
18.34).

Jumbo Shopping Bags

Red Retrospot, Pink Polkadot,
Suki variants. (Confidence:
65%+).

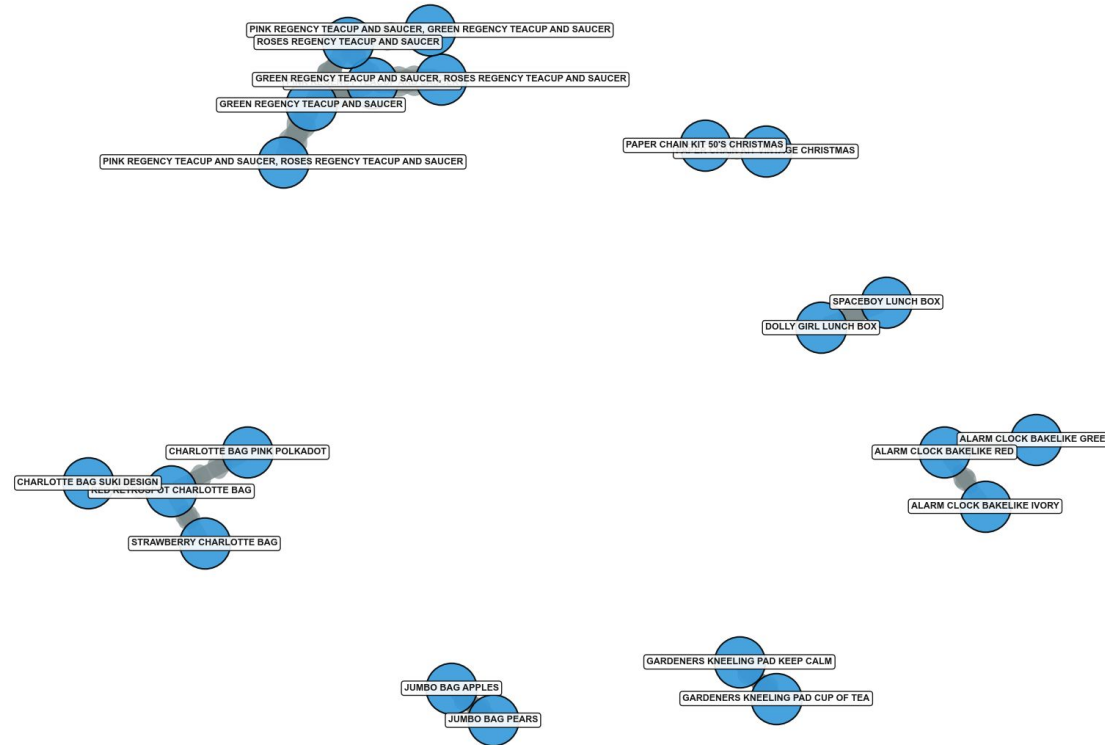
Garden Accessories

Kneeling pad variants. (Lift:
15.03).

Insight: Customers shop by theme and collection, not by random selection.

The Product Ecosystem

Product Association Network Graph (Top 30 Rules)



Key Takeaway: The network organically groups into dense clusters (Tea sets, Bags, Clocks), providing a literal roadmap for physical store layout optimization.

Decoding Customer Behavior



Category Loyalty:

Customers consistently purchase multiple variations of the same product line in a single trip.



Themed Baskets:

High confidence rules prove transactions are heavily themed (e.g., kitchenware vs. garden items).



Predictability:

These patterns are mathematically predictable, making them the perfect foundation for recommendation engines.

Strategic Recommendations (Digital & Marketing)



Algorithmic Cross-Selling:

Inject "Frequently Bought Together" prompts at digital checkout (e.g., If Pink Polkadot bag is in cart, trigger Red Retrospot).



Targeted Promotions:

Launch "Mix-and-Match" campaigns specifically targeting the Jumbo Bag cluster to drive volume.

Strategic Recommendations (Physical Retail)



Hard Bundling:

Package the Regency Tea Sets as a single cohesive SKU at a slight discount to guarantee high-volume turnover.



Layout Optimization:

Place highly associated items (like the Garden Kneeling Pads) on the same end-cap to eliminate in-store purchase friction.

Projected Business Impact

1. Revenue Growth:

Increased Average Order Value (AOV) via strategic cross-selling.

2. Customer Experience:

Reduced decision fatigue through personalized recommendations.

3. Supply Chain:

Predictive inventory planning for high-support anchor items to prevent cascading stockouts.

4. Marketing ROI:

Transitioning from intuition-based sales to mathematically proven campaigns.

SUMMARY

Successfully processed 522,064 raw records into 17,597 optimized baskets, deploying Machine Learning to map exact consumer purchasing behaviors.

RESULT

The business is now equipped to execute data-driven cross-selling, optimized merchandising, and highly profitable product bundling.